XIAMEN UNIVERSITY MALAYSIA

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Course Name	:	Regression Analysis
Lecturer	:	Lim Min
Academic Session	:	2025/04
Assessment Title	:	Project 1: Simple Linear Regression
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Prepared by	:	
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Feedback from Lecturer :

MARKING RUBRICS

Component Title	Report				Total Marks	100
Criteria	Score and Descriptors					Marks
	Excellent	Good	Average	Need Improvement	Poor	
Introduction	10	8	6	4	2	
Data Analysis - exploratory	10	8	6	4	2	
Data Analysis - formal test	10	8	6	4	2	
Data Analysis - validation	10	8	6	4	2	
Data Analysis - predictions	10	8	6	4	2	
Conclusions	10	8	6	4	2	
Recommendations	10	8	6	4	2	
R Code and R Output	10	8	6	4	2	
Use of language is appropriate, with no grammatical errors.	10	8	6	4	2	
Overall work is neat, well-organised and creative.	10	8	6	4	2	
					TOTAL	

Own Work Declaration

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I confirm that all sources cited in this work have been correctly acknowledged and that I fully understand the serious consequences of any intentional or unintentional academic misconduct.

In addition, I affirm that this submission does not contain any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. This work is my original creation, and it has not been based on any work of other students (past or present), nor has it been previously submitted to any other course or institution.

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1. Introduction

The purpose of this project is to develop a simple linear regression model to predict the log-transformed price of used cars based on their mileage driven, using data from a certain car brand. The analysis aims to explore and quantify the relationship between driven mileage and car price, providing a statistical basis for understanding vehicle depreciation.

The dataset used in this study consists of 200 observations, each representing a different used car of a certain brand and their price. Two main variables are used in the analysis:

1. Price: The selling price of the used car in Euros. Since car prices tend to vary exponentially instead of linearly with mileage, the price variable is log-transformed (log(price)) to better meet the assumptions of linear regression.

2. Mileage: The total mileage (miles) of the car has been driven. This serves as the sole regressor variable in the model.

For the purpose of model fitting (estimation), the dataset is split into two set: estimation data (training data, 70%), and prediction data (test data, 30%), and use the estimation data to train the regression model. The prediction data is used to validate the performance of the model.

To achieve the objective of understanding how mileage affects the price of a used car, a series of ten structured analytical tasks was conducted. First, scatter plots were generated to visually inspect the relationship between mileage and both raw price and log-transformed price (log(price)), providing an initial insight into whether linear modeling is appropriate. A sample correlation coefficient between mileage and log(price) was then calculated using the estimation data to quantify the strength and direction of the relationship.

Besides, a simple linear regression model was fitted using log(price) as the response variable and mileage as the regressor, then print the summary output to get basic statistical view. The fitted model was used to make predictions on a separate set of test (prediction) data, where the predicted prices were compared visually to actual prices through a third scatter plot.

Subsequently, statistical inference was performed: 95% confidence intervals were computed for the mean predicted price of cars driven 9,446 and 18,513 miles respectively, and 95% prediction intervals were calculated for the price of individual cars with the same mileage values.

These tasks offered both an exploratory and inferential understanding of how mileage impacts used car prices, and how the model can be used for future price predictions of used car and decision-making.

2. Data Analysis

2.1. Mileage vs Price.

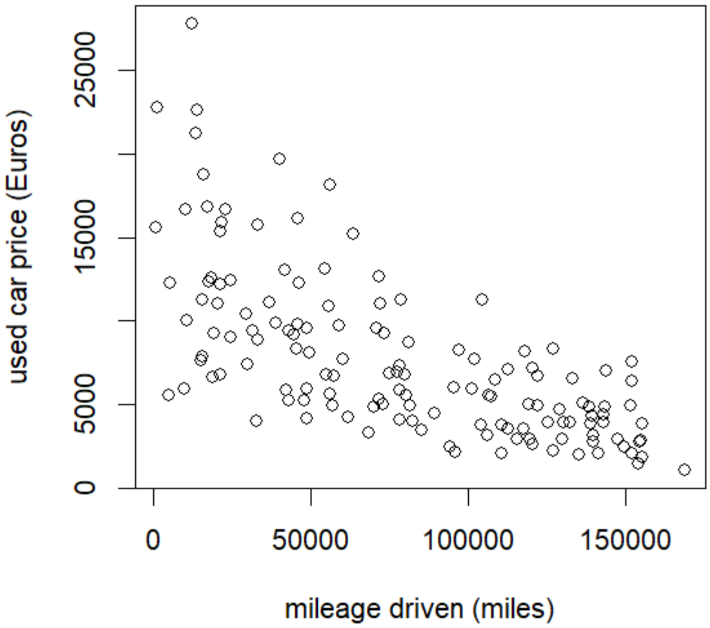


Figure 1. scatter diagram for mileage (miles) and price (Euros) using the estimation data.

The figure 1 displays a scatter diagram illustrating the relationship between mileage driven (miles) and used car price (Euros), based on the estimation dataset. Each point represents a used car, showing how its price varies with the number of miles it has been driven.

From the plot, we observe a clear **negative relationship**: as the mileage increases, the used car price generally decreases. However, this relationship is **not linear**. The decline in price is steep at low mileage levels (left half of the figure), but the rate of decrease flattens as mileage increases (right half of the figure). This pattern suggests a non-linear relationship, indicating that a linear model may not be appropriate for the untransformed price data. In particular, the spread of the prices becomes narrower and the variability decreases as mileage increases, which also violates the assumption of constant variance required for linear regression.

To address this issue, a transformation of the response variable is typically applied to linearize the relationship and stabilize the variance in next step.

2.2. **Mileage vs log(price).**

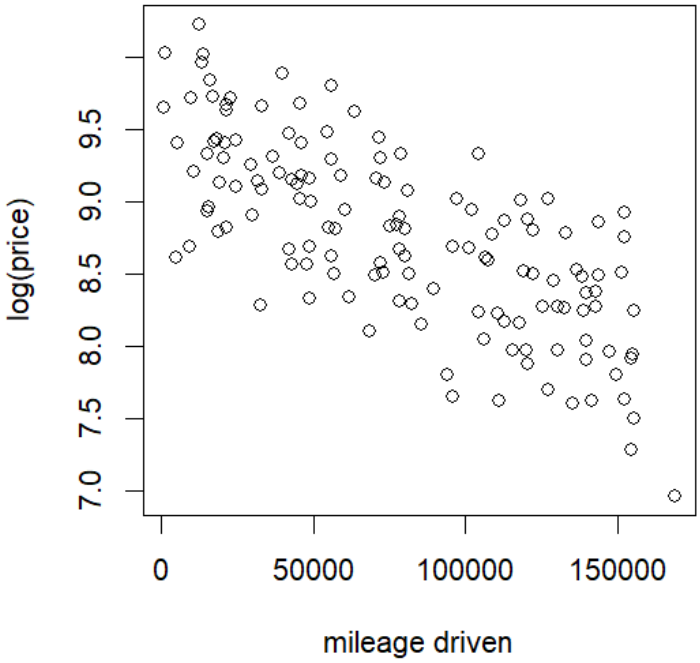


FIGURE 2. scatter diagram for mileage (miles) and log(price) (Euros) using the estimation data.

The figure 2 presents a scatter diagram showing the relationship between log(price) and mileage driven, based on the estimation dataset. Unlike the figure 1, the response variable (used car price) has been transformed using the logarithm.

This transformation has significantly improved the linearity of the relationship. We now observe a clear, approximately linear downward trend: as the mileage increases, the log of the used car price tends to decrease steadily. The spread of the points is also more uniform across the range of mileage, suggesting constant variance, which is a key assumption of linear regression models.

Due to this linearity and stabilized spread, a simple linear regression model is now appropriate for modeling the relationship between mileage and log(price). This conclusion is further supported by the sample correlation coefficient, which was calculated to be $= -0.7303859$, indicating that there is a negative moderate linear relationship between mileage (miles) and $\log(price)$, reinforcing the suitability of a straight-line model.

2.3. **Simple Linear Regression Model.**

Next, a simple linear regression model is fitted to the estimation data with log(price) as the response and mileage as the regressor, then print the summary output:

	Estimate	**Std. Error**	t	P**-value**
(Intercept)	9.535	7.151×10^{-2}	133.33	$< 2 \times 10^{-16}$
mileage	-9.825×10^{-6}	7.821×10^{-7}	-12.56	$< 2 \times 10^{-16}$

$R^2 = 0.5335$ Residual Standard Error $= 0.4345$ on df $= 138$

F-statistic: 157.8 on 1 and 138 df p-value: $< 2.2 \times 10^{-16}$

Table 1. Summary Output

Table 1 shows that the following information.

We consider the following simple linear regression model:

$$\log(price) = \beta_0 + \beta_1 mileage + \epsilon$$

where $\log(price)$ is the natural logarithm of the used car price (response), mileage is the number of miles the car has been driven (regressor), ϵ is the random error term assumed to be normally distributed with mean 0 and constant variance.

Hypothesis test to determine if the regression relation is significant: $H_0 : \beta_1 = 0$ $H_1 : \beta_1 \neq 0$ From the output: p-value $< 2 \times 10^{-16}$. Since the p-value is far less than 0.05, we reject H_0 and conclude that there is a statistically significant linear relationship between mileage and $\log(price)$.

The multiple R^2 is 0.5335. This means that approximately 53.35% of the variation in $\log(price)$ can be explained by the linear relationship with mileage.

Fitted function:

$$\log(\hat{price}) = \hat{\beta}_0 + \hat{\beta}_1 mileage$$

$$\log(\hat{price}) = 9.535 - 9.825 \times 10^{-6} mileage$$

Or, to express the price directly:

$$\hat{price} = \exp\left(9.535 - 9.825 \times 10^{-6} mileage\right)$$

Intercept $\beta_0 = 9.535$:

When the mileage is 0 (a new car), the expected log(price) is 9.535, which corresponds to an estimated price of $\exp(9.535) = 13836$ Euros.

Slope $\beta_1 = -9.825 \times 10^{-6}$

For every additional 1 mile driven, the expected log(price) decreases by 9.825×10^{-6}. However, in terms of price, the rate of decrease is not constant since it is not a linear relationship. As shown in Table 2, the price drops significantly with increased mileage.

TABLE 2. Predicted used car prices for selected mileage values

Mileage (miles)	**Predicted Price (Euros)**
100	13,819.21
1,000	13,697.55
10,000	12,538.34
100,000	5,178.58
1,000,000	0.75

2.4. **Predict price by using the fitted function.**

Next, the fitted function obtained previously is used to predict the price. Then, producing a scatter diagram for the actual price from the prediction data (test set) and prediction price from model.

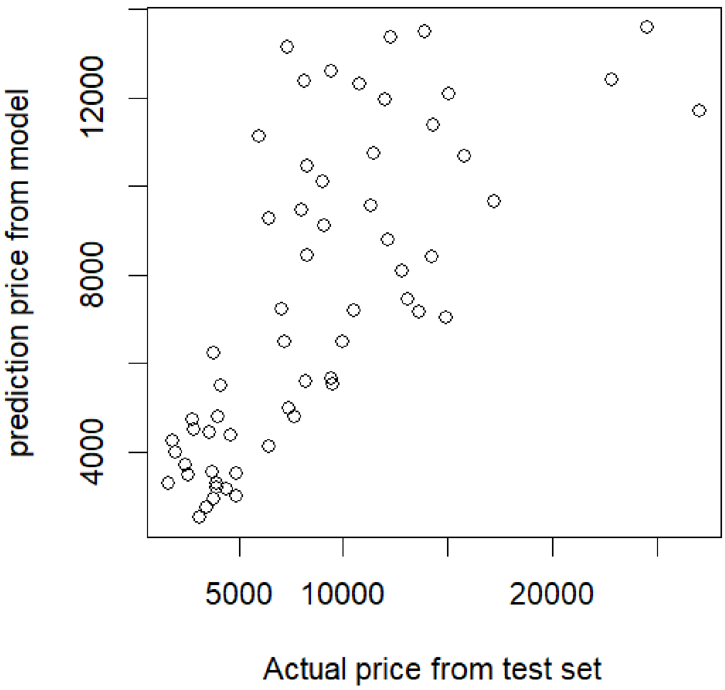


FIGURE 3. scatter diagram for actual price from prediction data (test set) and price_p (prediction price from model).

Using the fitted function, we compute the predicted log(price) for the cars in the prediction dataset, using the predict() function in R. However, since the model predicts log(price), we apply the exponential function to transform it back to price.

The predicted price from model can be compared against the actual prices in the prediction dataset. This helps to validate the performance of the model. Validate the model by plotting a scatter diagram of actual vs predicted prices. If the points lie close to the line $y = x$. It suggests good prediction accuracy, else the model may not be fitting well.

2.5. **Confidence and Prediction Intervals.**
A confidence interval estimates the range in which the mean of the dependent variable is likely to fall, given a specific value of the independent variable. It reflects the uncertainty around the regression line. In contrast, a prediction interval estimates the range where a new individual observation is likely to fall. It accounts for both the uncertainty in the mean estimate and the variability of individual data points. Table 3 shows two confidence intervals and prediction intervals of different kind of mileage.

Mileage	**Interval Type**	**Fit**	**Lower Bound**	**Upper Bound**	**Width**
9,446	Confidence	€12,606.77	€11,080.08	€14,343.83	€3,263.75
	Prediction	€12,606.77	€5,287.76	€30,056.33	€24,768.57
18,513	Confidence	€11,532.28	€10,251.18	€12,973.47	€2,722.29
	Prediction	€11,532.28	€4,844.87	€27,450.33	€22,605.46

Table 3. 95% Confidence and Prediction Intervals for Price (Including Point Estimates)

The confidence intervals are much narrower than the prediction intervals at both mileage 9446 and 18513. This is expected and reasonable because confidence intervals estimate the mean price, where prediction intervals account for the variability in individual car prices. As mileage increases, both intervals become slightly narrower, indicating a more stable prediction for higher mileage vehicles.

3. Conclusions and Recommendations

3.1. **Conclusion.**

A simple linear regression was calculated to predict the used car price of a certain brand based on its mileage driven. A significant regression equation was found ($t(138) = -12.56$, $p < 2 \times 10^{-16}$), with $R^2 = 0.5335$. The regression equation was:

$$\log(\hat{price}) = 9.535 - 9.825 \times 10^{-6} mileage$$

This indicates that for every 1-mile increase in mileage, the log of predicted price decreases by 9.825×10^{-6}. However, in terms of price, the rate of decrease is not constant since it is not a linear relationship. The price drops significantly with increased mileage. The model explains approximately 53.35% of the variability in the log of car prices.

3.2. **Strengths of the Analysis.**

Statistically significant finds the relationship between mileage and log(price) is highly significant (p ¡ 0.001), indicating a strong negative association between mileage and car price within the sample.

With an $\mathbf{R^2}$ value of 0.5335, the model explains over 53% of the variance in log(price) using only one regressor variable. This is a strong result for a simple linear regression model and shows that mileage is a powerful regressor of used car value.

Transforming the response variable (price) using the logarithm helps to satisfy linear regression assumptions, particularly linearity. This improves both interpretability and model reliability.

3.3. **Weaknesses and Limitations.**

The model considers only mileage as a regressor. Other influential factors such as registration year, price (Euros), transmission type of gear box (auto, manual), fuel type (petrol, hybrid, others), road tax (Euros), miles per gallon, and engine size (litres), and accident history are not included. Ignoring these may lead to biased or incomplete results.

Assumption violations not tested in-depth. The analysis does not deeply test key regression assumptions such as normality of residuals and independence. If these assumptions are violated, it may affect the validity of the inference.

Since the data focuses on a specific car brand, the findings may not generalize well to other brands, vehicle types, or regions. Meanwhile, the sample size may not be large enough since it only cover 200 samples.

Log-transformed dependent variable complicates the interpretation. Although the log transformation improves model performance, but it also becomes complex in interpreting the results directly in terms of dollars, especially for audiences without statistical knowledge.

3.4. **Recommendations.**

Including more relevant predictors can be one of the methods to improve the model. We can incorporate additional variables such as car age, engine type, transmission, fuel efficiency, and brand/model-specific effects to improve the model's accuracy and address omitted variable bias.

Conduct some formal tests, such as residual plots, Q-Q plots, to ensure that linear regression assumptions are met.

To enhance applicability, consider conducting separate regressions for different vehicle categories, such as sedans, bus, truck or brands, as the relationship between mileage and price might vary substantially.

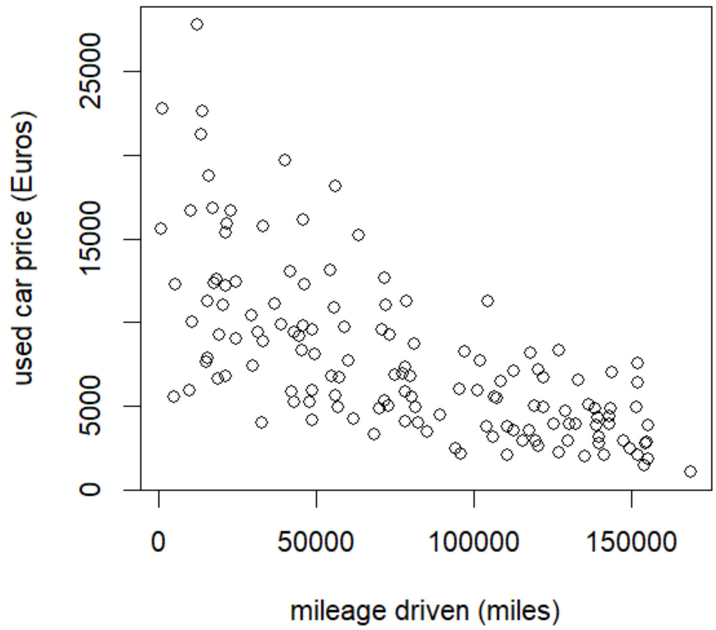
Provide back-transformed results (exponentiate the fitted log values) when communicating with general audiences who do not have statistical knowledge to make price predictions more intuitive.

We can explore non-linear relationships or machine learning methods (such as random forest or gradient boosting) since the true relation between mileage and price is more complex than a straight line can capture.

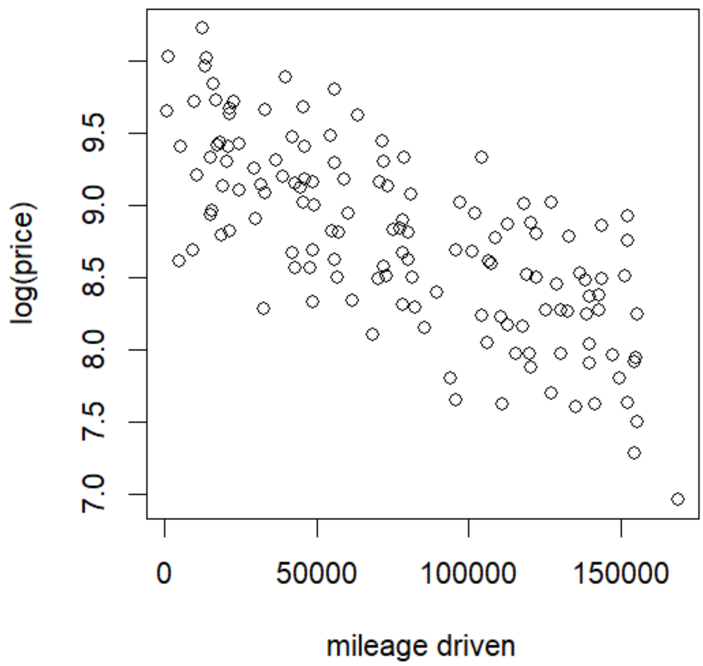
4. R code and R output

```
> CarData <- read.table("C:/University/Year 3 Sem 1/Regression Analysis/DSC2304093.txt",
    header = TRUE)
> set.seed(0611)
> dt <- sort(sample(nrow(CarData), nrow(CarData)* 0.7))
> CarData_e <- CarData[dt, ] #estimation
> CarData_p <- CarData[-dt, ] #prediction

> plot(CarData_e $mileage, CarData_e $price, xlab = "mileage driven (miles)" ,
    ylab = "used car price (Euros)")
```




```
> plot(CarData_e $mileage, log(CarData_e $price), xlab = "mileage driven" , ylab = "log(price)")
```

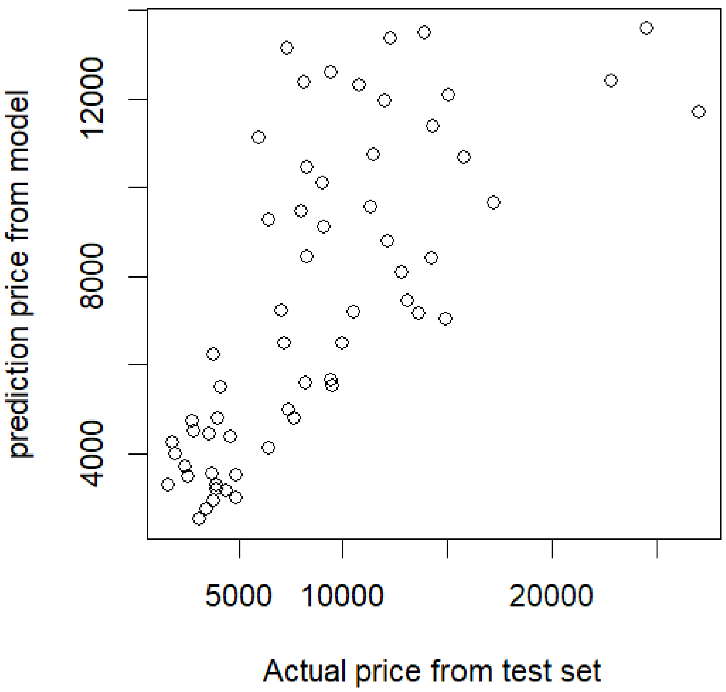


```
> cor(CarData_e $mileage, log(CarData_e $price))
[1] -0.7303859
> CarData_e.model <- lm(log(price) ~ mileage, data = CarData_e)
> summary(CarData_e.model)
Call:
lm(formula = log(price) ~ mileage, data = CarData_e)

Residuals:
     Min       1Q   Median       3Q      Max
-0.94000 -0.30955  0.03235  0.32927  0.88914

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.535e+00  7.151e-02  133.33   <2e-16 ***
mileage     -9.825e-06  7.821e-07  -12.56   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4345 on 138 degrees of freedom
Multiple R-squared:  0.5335, Adjusted R-squared:  0.5301
F-statistic: 157.8 on 1 and 138 DF,  p-value: < 2.2e-16
> log_price_p <- predict(CarData_e.model, newdata = CarData_p)
> price_p <- exp(log_price_p) # Because the model is log(price), we need to use exp()
    to restore it to price
>
> plot(CarData_p $price, price_p, xlab = "Actual price from test set",
    ylab = "prediction price from model")
```



```
> #7
> conf_9446 <- predict(CarData_e.model, newdata = data.frame(mileage = 9446),
    interval = "confidence", level = 0.95)
> exp(conf_9446)
       fit      lwr      upr
1 12606.77 11080.08 14343.83

> #8
> conf_18513 <- predict(CarData_e.model, newdata = data.frame(mileage = 18513),
    interval = "confidence", level = 0.95)
> exp(conf_18513)
       fit      lwr      upr
1 11532.28 10251.18 12973.47
>
> #9
> pred_9446 <- predict(CarData_e.model, newdata = data.frame(mileage = 9446),
    interval = "prediction", level = 0.95)
> exp(pred_9446)
       fit     lwr      upr
1 12606.77 5287.76 30056.33
>
> #10
> pred_18513 <- predict(CarData_e.model, newdata = data.frame(mileage = 18513),
    interval = "prediction", level = 0.95)
> exp(pred_18513)
       fit      lwr      upr
1 11532.28 4844.874 27450.33


> #extra 1
> mileages <- c(100, 1000, 10000, 100000, 1000000)
> new_mileage_df <- data.frame(mileage = mileages)
>
> # Predict log(price)
> log_price_preds <- predict(CarData_e.model, newdata = new_mileage_df)
>
> # Convert back to price
> price_preds <- exp(log_price_preds)
>
> # Result table
> data.frame(mileage = mileages, predicted_price = round(price_preds, 2))
  mileage predicted_price
1   1e+02        13819.21
2   1e+03        13697.55
3   1e+04        12538.34
4   1e+05         5178.58
5   1e+06            0.75
```